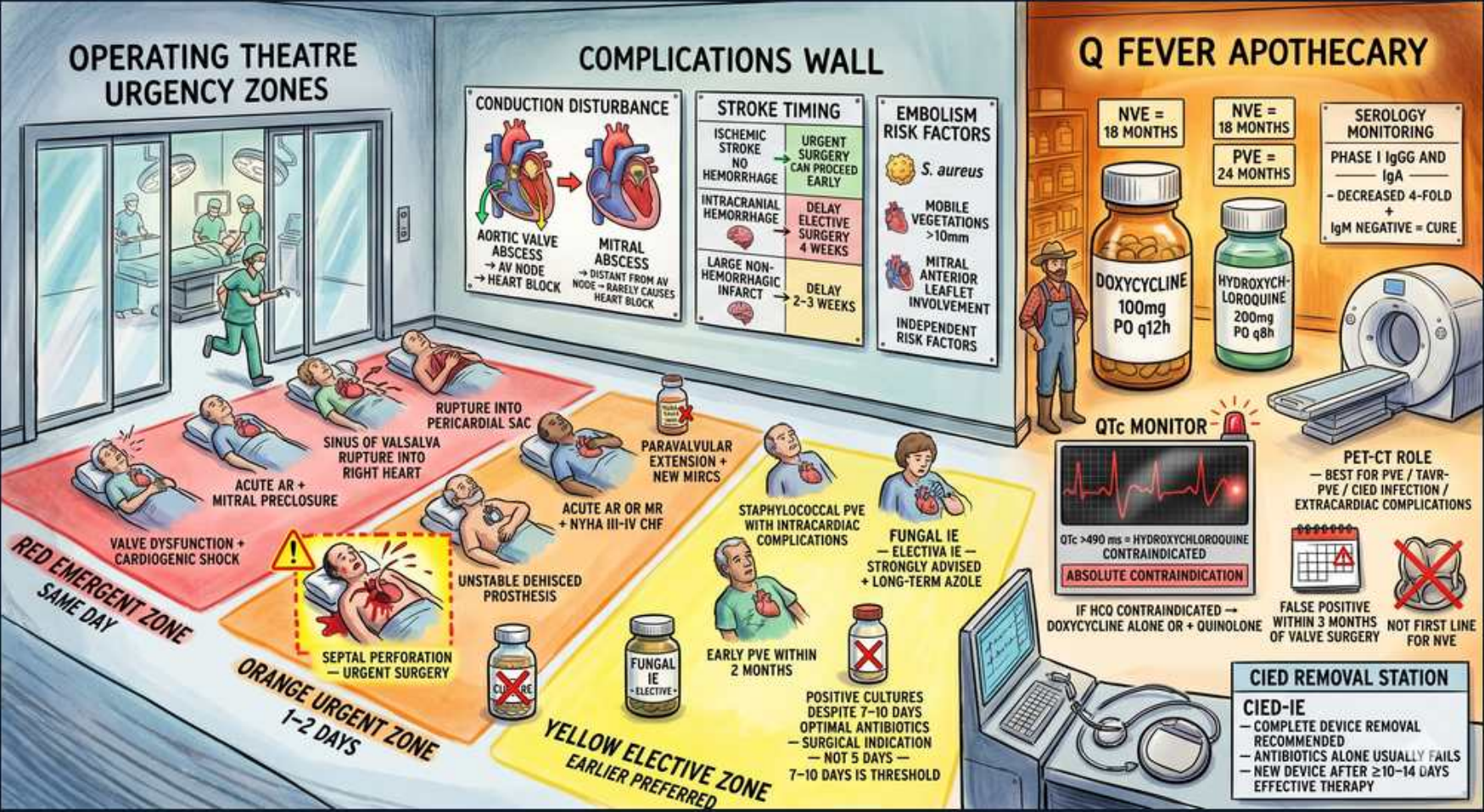


Infective Endocarditis — Complications & Surgical Urgency



1. Aortic abscess → heart block

WHAT YOU SEE

Conduction disturbance: aortic root abscess, AV block

WHAT IT ENCODES

An aortic-root/perivalvular abscess can extend into the conduction system, causing progressive AV block. New PR prolongation or AV block on serial ECG signals abscess and is a surgical indication.

BOARD TRAP

New AV block in IE = perivalvular abscess until disproven — get TEE; don't just place a pacemaker.

2. Mitral abscess

WHAT YOU SEE

Mitral abscess / conduction box

WHAT IT ENCODES

Mitral annular abscess is another perivalvular extension; abscesses overall demand surgery because antibiotics cannot sterilize the cavity. TEE is the imaging test of choice.

BOARD TRAP

Abscess is a structural surgical problem — prolonged antibiotics alone won't cure it.

3. Stroke timing — ischemic OK

WHAT YOU SEE

Stroke timing: ischemic non-hemorrhagic, surgery not delayed

WHAT IT ENCODES

After a small/asymptomatic ischemic embolic stroke, valve surgery need not be delayed if indicated. Large infarcts or hemorrhagic transformation push delay 2-4 weeks due to bypass heparinization.

BOARD TRAP

A small ischemic stroke does NOT mandate delaying surgery — only hemorrhage/large infarct does.

4. Hemorrhage — delay 2-4 wks

WHAT YOU SEE

Intracranial hemorrhage, delay elective surgery 2-4 weeks

WHAT IT ENCODES

Intracranial hemorrhage requires delaying cardiac surgery roughly 2-4 weeks because cardiopulmonary bypass heparin risks fatal hematoma expansion.

BOARD TRAP

The delay rule is for HEMORRHAGE — don't apply it to every embolic event.

5. Embolism risk factors

WHAT YOU SEE

Embolism risk: *S. aureus*, large/mobile vegetation, mitral, anterior leaflet

WHAT IT ENCODES

Highest embolic risk: vegetation >10 mm, mobile, mitral (especially anterior leaflet) location, and *S. aureus*. Embolic risk is highest in the first 2 weeks and falls sharply with effective antibiotics.

BOARD TRAP

Embolic risk drops fast once antibiotics start — prophylactic surgery for vegetation size alone is nuanced, weigh recurrent emboli.

6. Red zone — emergent same day

WHAT YOU SEE

Red emergency zone: valve dysfunction, cardiogenic shock, acute AR/MR, same day

WHAT IT ENCODES

Emergent (same-day) surgery: refractory pulmonary edema or cardiogenic shock from acute severe AR/MR, or sinus of Valsalva rupture into a chamber/pericardium causing tamponade.

BOARD TRAP

Hemodynamic collapse trumps everything — operate same-day even mid-antibiotic course.

7. Sinus of Valsalva / pericardial rupture

WHAT YOU SEE

Rupture into pericardial sac / right atrium

WHAT IT ENCODES

Aortic-root infection can rupture into the pericardium (tamponade) or a cardiac chamber (acute shunt/HF) — both are catastrophic, emergent surgical emergencies.

BOARD TRAP

Sudden hemodynamic decompensation with a new continuous murmur may be a fistula/rupture — image and operate emergently.

8. Orange zone — urgent 1-2 days

WHAT YOU SEE

Orange urgent zone: unstable dehiscence prosthesis, septal abscess

WHAT IT ENCODES

Urgent (1-2 day) surgery: unstable/dehiscence prosthesis, perivalvular abscess with conduction disturbance, persistent infection, or large vegetation with recurrent emboli.

BOARD TRAP

A dehiscence prosthetic valve is urgent — not something to manage medically while awaiting culture clearance.

9. Septal abscess

WHAT YOU SEE

Septal abscess / urgent surgery placard

WHAT IT ENCODES

Interventricular septal or perivalvular abscess is an urgent surgical indication; ongoing destruction risks heart block, fistula, and embolization.

BOARD TRAP

Abscess + new conduction disease should not wait — urgent operative source control.

10. Staph + intracardiac complication

WHAT YOU SEE

Staphylococcal PVE with intracardiac complications

WHAT IT ENCODES

S. aureus prosthetic-valve IE, especially with intracardiac complications (abscess, dehiscence), has very high mortality and usually requires surgery in addition to vancomycin/dapto + rifampin + gentamicin.

BOARD TRAP

Staph PVE is a surgical disease — antibiotics alone frequently fail.

11. Fungal IE — surgery + azole

WHAT YOU SEE

Fungal IE: strongly advised long-term azole, +/- echinocandin

WHAT IT ENCODES

Fungal IE (*Candida*, *Aspergillus*) almost always needs valve surgery plus prolonged antifungals (echinocandin/amphotericin then long-term azole suppression), as medical cure alone is rare.

BOARD TRAP

Fungal IE treated medically only is doomed — surgery is part of cure, with long-term azole suppression.

12. Yellow zone — elective

WHAT YOU SEE

Yellow elective zone: earlier preferred

WHAT IT ENCODES

Elective (earlier-preferred) surgery: controlled infection with structural indication such as severe regurgitation without HF, healed but significant valve damage. Earlier operation generally improves outcomes.

BOARD TRAP

Even 'elective' IE surgery should not be deferred long — earlier intervention improves survival.

13. Persistent positive cultures 7-10 d

WHAT YOU SEE

Positive cultures despite 7-10 days / surgical indication

WHAT IT ENCODES

Persistently positive blood cultures despite 7-10 days of appropriate antibiotics indicate uncontrolled focus (abscess, resistant organism) and constitute a surgical indication.

BOARD TRAP

Persistent bacteremia after a week of correct therapy means source control is needed — not just longer antibiotics.

14. Q fever apothecary — PVE doxy

WHAT YOU SEE

Q fever: NVE 18 months, PVE 24 months doxycycline+HCQ

WHAT IT ENCODES

Chronic Q fever IE: doxycycline + hydroxychloroquine, 18 months for native and 24 months for prosthetic valves, with serologic monitoring (target phase I IgG decline ≥ 4 -fold).

BOARD TRAP

Q fever IE needs years of therapy with serologic monitoring — not a standard 4-6 week course.

15. PET-CT for PVE

WHAT YOU SEE

PET-CT role: best for PVE, false-positive early, not first-line NVE

WHAT IT ENCODES

FDG-PET/CT aids prosthetic-valve IE diagnosis and finds extracardiac emboli but gives false positives within ~3 months of surgery and is not first-line for native valves.

BOARD TRAP

PET-CT in the early post-op window is unreliable (inflammatory uptake) — interpret with caution.

16. CIED removal

WHAT YOU SEE

CIED removal station: complete device removal recommended

WHAT IT ENCODES

Cardiac implantable electronic device (pacemaker/ICD) infection mandates COMPLETE hardware removal (leads + generator) plus antibiotics — antibiotics alone almost always fail.

BOARD TRAP

You cannot salvage an infected CIED with antibiotics alone — the entire system must come out.